YOUNG'S DOUBLE-SLIT INTERFERENCE

1. Definition

Young's double-slit experiment demonstrates the wave nature of light through the phenomenon of interference. It was performed by Thomas Young in 1801.

Interference is the phenomenon in which two or more coherent light waves superpose to produce a redistribution of light intensity, forming alternate bright and dark fringes on a screen.

2. Experimental Setup

The experiment consists of the following components:

1. A monochromatic light source that produces light of a single wavelength.
2. A narrow slit that provides light to two closely spaced slits.
3. Two narrow slits, S_1 and S_2, separated by a distance d.
4. A screen placed at a distance D from the double slits, where D is much greater than d.

The two slits act as coherent sources of light waves. When these waves overlap on the screen, an interference pattern is formed.

3. Principle of Interference

The interference pattern is produced due to the superposition of light waves from the two coherent slits.

There are two types of interference:

1. Constructive Interference

When two light waves meet in the same phase, their amplitudes add together, producing a bright fringe.

Condition:

Path difference = $n\lambda$

2. Destructive Interference

When two light waves meet in opposite phases, their amplitudes cancel each other, producing a dark fringe.

Condition:

Path difference = $(2n + 1)\lambda/2$

Where:

- n = Order of the fringe (integer)
- λ = Wavelength of light

4. Path Difference

The path difference is the difference between the distances travelled by the two light waves to reach a point on the screen.

If the slit separation is d and the observation angle is θ, the path difference is:

$\Delta = d \sin \theta$

For small angles:

$\sin \theta \approx \tan \theta \approx y/D$

Therefore,

$\Delta = dy/D$

Where:

- d = Distance between the two slits
- D = Distance between the slits and the screen
- y = Distance of the observation point from the central fringe

5. Position of Bright Fringes

Bright fringes are formed due to constructive interference.

Condition:

$\Delta = n\lambda$

Substituting $\Delta = dy/D$:

$dy/D = n\lambda$

Therefore, the position of the nth bright fringe is:

$y_n = n\lambda D/d$

Where n = 0, ±1, ±2, ±3, ...

For the central bright fringe, n = 0:

$y_o = 0$

Hence, the central fringe is bright.

6. Position of Dark Fringes

Dark fringes are formed due to destructive interference.

Condition:

$\Delta = (2n + 1)\lambda/2$

Substituting $\Delta = dy/D$:

$dy/D = (2n + 1)\lambda/2$

Therefore, the position of the nth dark fringe is:

$y_n = (2n + 1)\lambda D/(2d)$

Where n = 0, 1, 2, 3, ...

7. Fringe Width

Fringe width is the distance between two consecutive bright fringes or two consecutive dark fringes.

It is represented by β.

For two consecutive bright fringes:

$\beta = y_{n+1} - y_n$

Substituting the positions of bright fringes:

$\beta = (n + 1)\lambda D/d - n\lambda D/d$

Therefore, the fringe width is:

$\beta = \lambda D/d$

Where:

- β = Fringe width
- λ = Wavelength of light
- D = Distance between the slits and the screen
- d = Distance between the two slits

8. Factors Affecting Fringe Width

1. Wavelength (λ): Fringe width increases when wavelength increases.
2. Screen distance (D): Fringe width increases when the screen is moved farther away.
3. Slit separation (d): Fringe width decreases when the distance between the slits increases.

Thus,

$\beta \propto \lambda$

$\beta \propto D$

$\beta \propto 1/d$

9. Characteristics of the Interference Pattern

1. Alternate bright and dark fringes are formed on the screen.
2. The central fringe is bright when the two coherent sources are in phase.
3. Bright fringes correspond to constructive interference.
4. Dark fringes correspond to destructive interference.
5. For ideal narrow slits and small angles, the fringes are equally spaced.

6. The fringe width is the same for consecutive bright fringes and consecutive dark fringes.
7. The experiment provides evidence for the wave nature of light.

10. Applications

1. Determination of the wavelength of light.
2. Measurement of small distances using interference.
3. Study of the wave nature of light.
4. Comparison of wavelengths of different light sources.
5. Study of coherent light and interference phenomena.

Conclusion

Young's double-slit experiment demonstrates the wave nature of light by producing alternate bright and dark fringes through interference. The positions of the fringes depend on the wavelength of light, the slit separation, and the distance between the slits and the screen. The fringe width is given by $\beta = \lambda D/d$.